



School of Computer Science and Statistics

Week 2 Assignment

CS7CS2 – Optimisation for Machine Learning

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General Guidelines

- You may discuss ideas with classmates; however, **all written work and code must be entirely your own**. Do not share code. All submissions will be checked for plagiarism.
- Reports must be typed and submitted as a **single PDF along with a single .ipynb or .py file via Blackboard**. Do not submit the report inside a zip file.
- **Important:** Marks are awarded primarily for explanation, reasoning, and interpretation, not for raw code or numerical output.
 - Code-based results must be explained.
 - Numerical results must be interpreted.
 - Statements such as “see code” or unexplained plots/numbers are insufficient.
- All plots must have labelled axes, readable text, and a clear written explanation of what they show.

Submission Rules

Submit:

- A short PDF report with codes, plots and brief explanations.
 - A .ipynb or .py file containing the executable Python code.
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Q1.

(I) Let $f(x) = x^4$.

- Compute $\frac{df}{dx}$ using SymPy.
- For $x \in [-2, 2]$, compute the exact derivative and a forward finite-difference approximation with $\delta = 0.01$. Plot both.
- Vary $\delta \in [0.001, 1]$. Compute and plot the mean absolute error (MAE).
- Implement gradient descent

$$x_{k+1} = x_k - \alpha f'(x_k).$$

Use $x_0 = 1$ and $\alpha = 0.05, 0.5, 1.2$.

- Plot x_k and $f(x_k)$ versus iteration and comment on convergence.

Q2.

(I) Let

$$f(x_0, x_1) = \frac{1}{2}(x_0^2 + 10x_1^2), \quad \nabla f = [x_0, 10x_1].$$

- (a) Plot contours of f over $[-2, 2] \times [-2, 2]$.
- (b) From $x^{(0)} = [1.5, 1.5]$, run gradient descent with $\alpha = 0.05$ and $\alpha = 0.2$. Overlay the paths and comment on the narrow valley effect.

(II) Let

$$f(x) = x^4 - 2x^2 + 0.1x.$$

- (a) Plot $f(x)$ for $x \in [-2, 2]$.
- (b) Run gradient descent from $x_0 = -1.5$ and $x_0 = 1.5$ ($\alpha = 0.05$). Compare the final solutions and explain the role of local minima.

Q3.

(I) Let $f(x) = x^2$, $f'(x) = 2x$.

- (a) Run gradient descent with $x_0 = 1$ for $\alpha = 0.1, 0.01, 1.01$.
- (b) Plot x_k and $f(x_k)$ versus iteration (include a log-scale plot). Explain why $\alpha = 1.01$ fails and why log-scale is useful.

(II) Let $f(x) = \gamma x^2$, $f'(x) = 2\gamma x$. Fix $x_0 = 1$, $\alpha = 0.1$.

- (a) Run gradient descent for $\gamma \in \{0.5, 1, 2, 5\}$.
- (b) Plot $f(x_k)$ versus iteration (log-scale) and explain the effect of γ .

(III) Let $f(x) = |x|$ with subgradient $g(x) = \text{sign}(x)$, $g(0) = 0$.

- (a) Run gradient descent with $x_0 = 1$, $\alpha = 0.1$, for 60 iterations.
- (b) Plot x_k and $f(x_k)$ versus iteration and describe the oscillatory behaviour near $x = 0$.

Q4.

Let

$$f(x_1, x_2) = x_1^2 + \gamma x_2^2, \quad \nabla f = [2x_1, 2\gamma x_2].$$

- (a) For $\gamma = 1$ and $\gamma = 4$, plot contours over $[-1.5, 1.5]^2$.
- (b) From $x^{(0)} = [1, 1]$, run gradient descent with $\alpha = 0.1$ and overlay the paths. Explain the behaviour.

Now consider the function

$$f(x_1, x_2) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2.$$

- (c) Plot contours over $[-2, 2] \times [-1, 3]$.
 - (d) Starting from $x^{(0)} = [-1.25, 0.5]$, run gradient descent with $\alpha = 0.001$ and $\alpha = 0.005$ (2000 iterations). Comment on convergence.
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